

Hardware-Aware Quantum Bayesian Learner Compression

Linking Human Belief Updating to NISQ Circuit Complexity

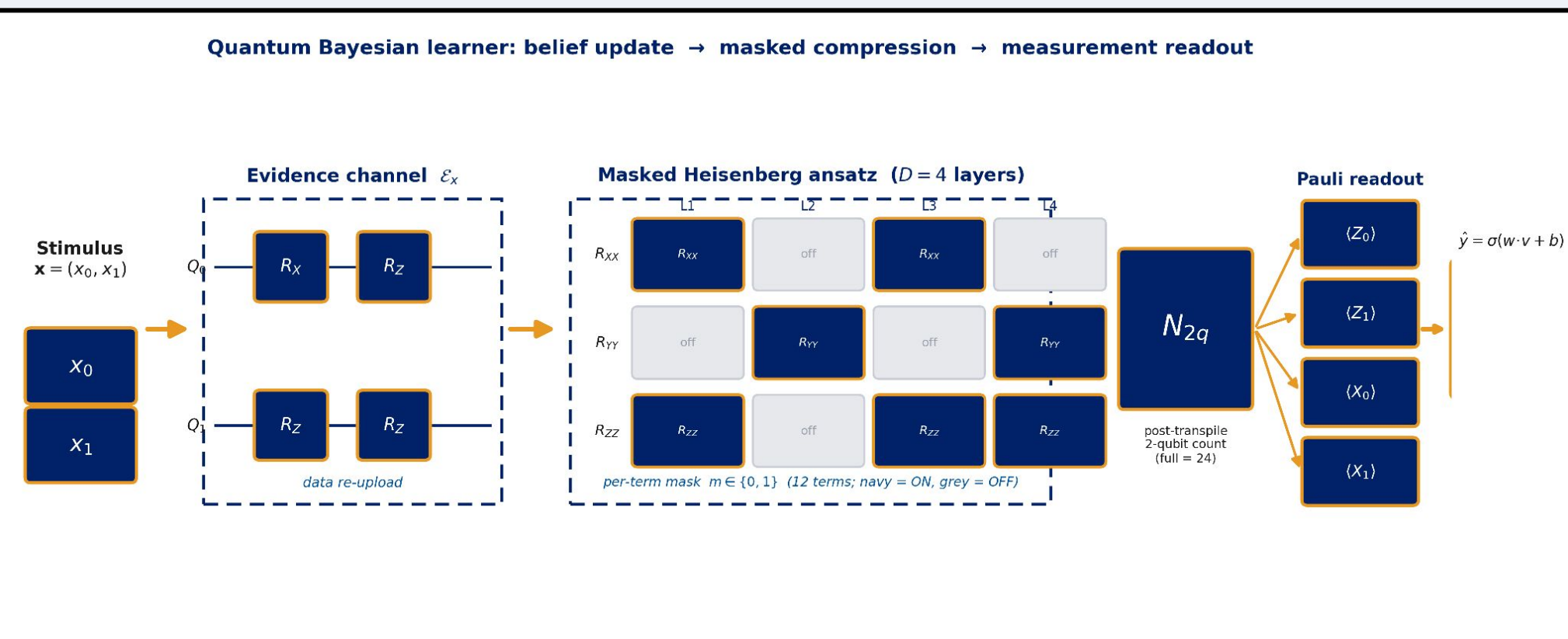
Mohammad Zoraiz

Duke University · Pratt School of Engineering · Durham, NC

1 Motivation — Two fields, one question

- Bayesian models cast human learning as **belief updating**: a prior over hypotheses, revised by evidence. **Quantum cognition** represents beliefs as density operators in Hilbert space, with evidence acting through **quantum channels**.
- On **NISQ** hardware, **two-qubit gates dominate the error budget**, so removing them without breaking the task is a central engineering goal.
- Central question**: *How much of a quantum Bayesian learner's entangling structure can we remove before it can no longer represent the task — and what happens on real hardware?*

2 The Quantum Bayesian Learner



- Belief = state ρ on a **2-qubit** register; prior ρ_0 = “no evidence yet.” A stimulus x updates the belief via a **CPTP evidence channel** $\mathcal{E}_x$ (ρ) = $\sum_i E_x \rho E_x^\dagger$.
- Update circuit = hardware-efficient ansatz: data re-uploading, per-layer single-qubit rotations (R_x , R_z), and masked Heisenberg interaction gates (R_{XX} , R_{YY} , R_{ZZ}). The interaction gates are transpiled to the backend basis. Depth **D=4**, with 2 re-uploads.
- Readout**: **single-qubit Pauli expectations** $v = (\langle Z_0 \rangle, \langle Z_1 \rangle, \langle X_0 \rangle, \langle X_1 \rangle)$ → trainable head $\hat{y} = \sigma(w \cdot v + b)$.
- Under this constrained readout, useful joint-feature structure appears only when some entangling terms remain.
- Two realizations, same conclusions: a fast **pure-state** model and a literal **mixed-state density matrix** (maximally mixed prior $I/2^n$, non-unital Lüders/POVM evidence filter + amplitude damping, Bayesian trace renormalization).

3 Method — mask, real cost, greedy pruning

- Per-term binary mask** $m_{\square, \square} \in \{0, 1\}$ over $p \in \{XX, YY, ZZ\}$ in each of $D=4$ layers ⇒ **12 maskable terms** ($R_{PP}(0)=I$ when off).
- Hardware-aware cost N_{2q}** = real post-transpile two-qubit gate count on IBM **FakeManila** (5-qubit, includes SWAPs), via Qiskit's transpiler. Full circuit ⇒ **$N_{2q} = 24$** (each active Heisenberg term = 2 CNOTs).
- Objective**: $L(\theta, m) = (1/N) \sum BCE(y_{\square}, \hat{y}) + \lambda \cdot N_{2q}(m)$.
- Greedy structured pruning**: from the full circuit, repeatedly drop the term whose removal least increases training cross-entropy, **warm-start retrain** → traces the accuracy-vs- N_{2q} **frontier** from 24 down to 0.
- Exact **JAX autodiff** (no parameter-shift); statevector verified **bit-for-bit against Qiskit** ($|1 - \text{fidelity}| \approx 4 \times 10^{-16}$).

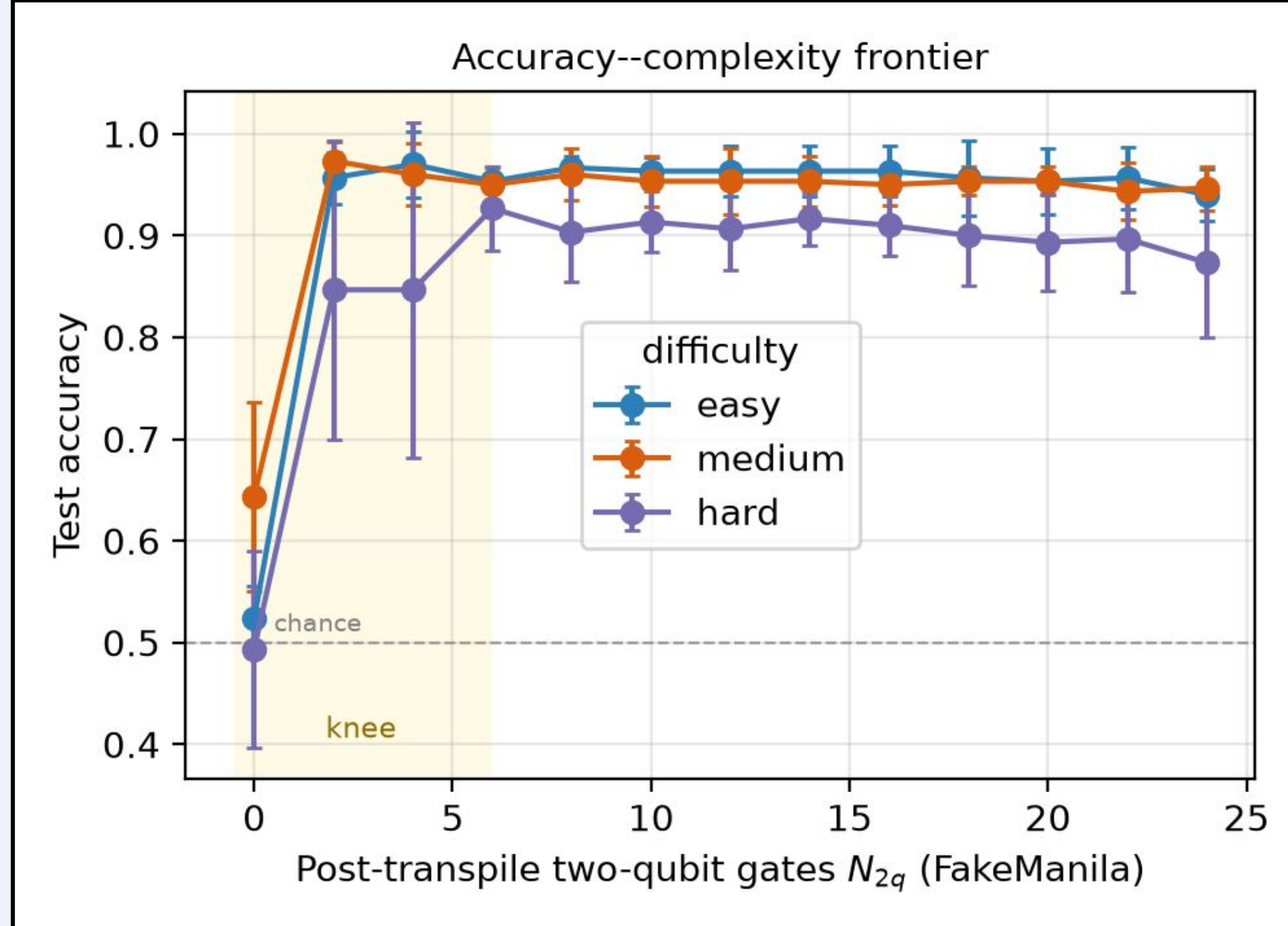
Task — controllable difficulty knob

- Checkerboard** categorization: $y = (L_f \cdot x_0 + L_f \cdot x_1) \bmod 2$, stimuli in $\{0, 1\}^2$.
- Difficulty f : **f=2 easy / f=3 medium / f=4 hard**. 200 stimuli, 70/30 split, balanced ⇒ **chance = 0.5**.
- Local similarity, but the rule depends on the **joint** features ⇒ needs the interaction only entanglement can supply.
- Same stimulus space, harder rule: difficulty changes through the category function rather than by changing dataset size or class balance.

Why it matters · Scope & limits

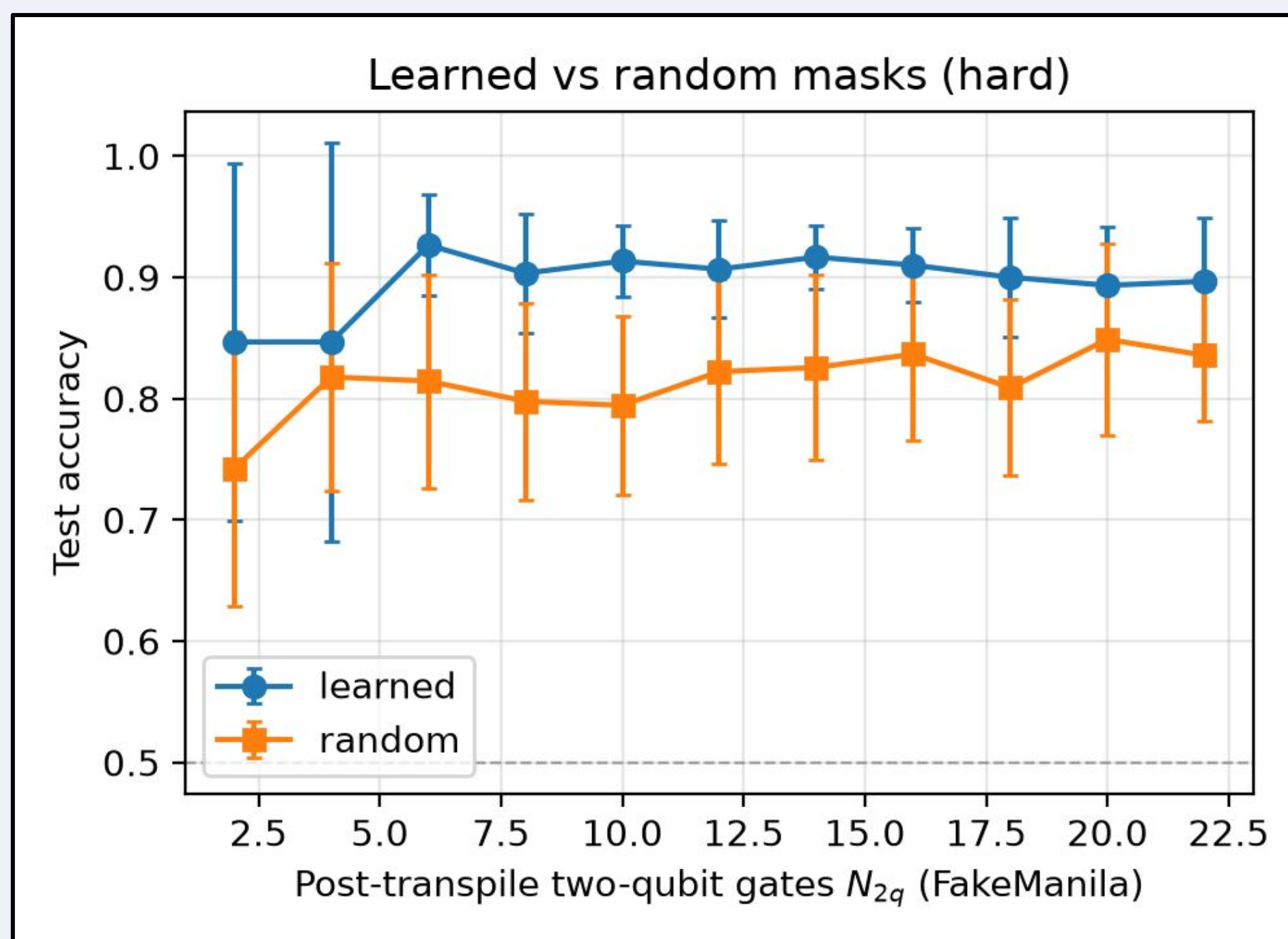
- Engineering relevance**: because N_{2q} is a **real post-transpile count**, this accuracy–cost knob applies to **any NISQ variational circuit** — independent of the cognitive framing.
- Controlled by design**: 2-qubit belief register, exact-simulated evidence channel, single-qubit-Pauli readout, and a synthetic task make every claim measurable and cheap to re-run.
- Limits**: hardware is a **single-seed demonstration** ($n = 16 / 40$ stimuli, defaults-only mitigation); N_{2q} is a FakeManila proxy, not the deployment device; density model uses 3 seeds.
- Not a claim about human cognition** — the cognitive language is an interpretive lens on a controlled circuit experiment.
- Because cost is measured after transpilation, the method tests deployable circuit complexity rather than symbolic circuit size.

4 Result 1 — The capacity boundary



- Above chance at all nonzero budgets (easy/medium $\approx 0.95\text{--}0.97$, hard $\approx 0.87\text{--}0.93$); frontier **nearly flat** → most entangling structure is **redundant**.
- At **$N_{2q} = 0$** easy/hard **collapse to chance** (0.52, 0.49); medium drops sharply (0.64) → some entanglement is **necessary**, but far less than the full circuit.
- Hard task: the optimum sits at or **below full cost** (0.93 @ $N_{2q}=6$ vs 0.87 @ $N_{2q}=24$; not significant at 5 seeds) → structured compression is **at worst free**.

5 Result 2 — Structure beats count

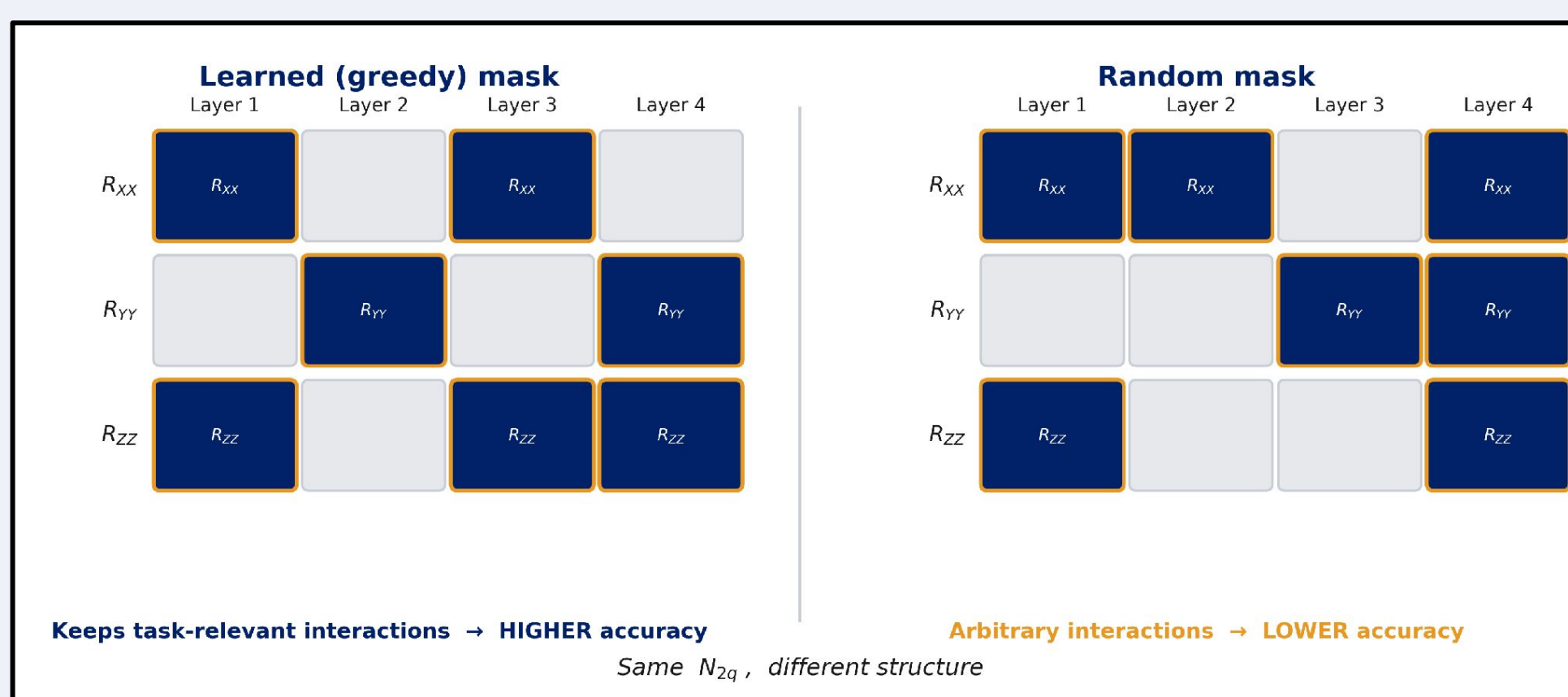


- Learned (greedy) masks vs **random masks at matched N_{2q}** .
- Learned wins at **every** budget by **~6–12 points**. Per seed: learned beats random in **49 / 55** paired comparisons, mean advantage **+0.083**, one-sided Wilcoxon **$p \approx 1.7 \times 10^{-6}$** . (*paired test; the unpaired bars above overlap by design*)
- ⇒ **Which entanglers you keep matters, not just how many.**

Hard task — learned vs random (mean acc, 5 seeds)

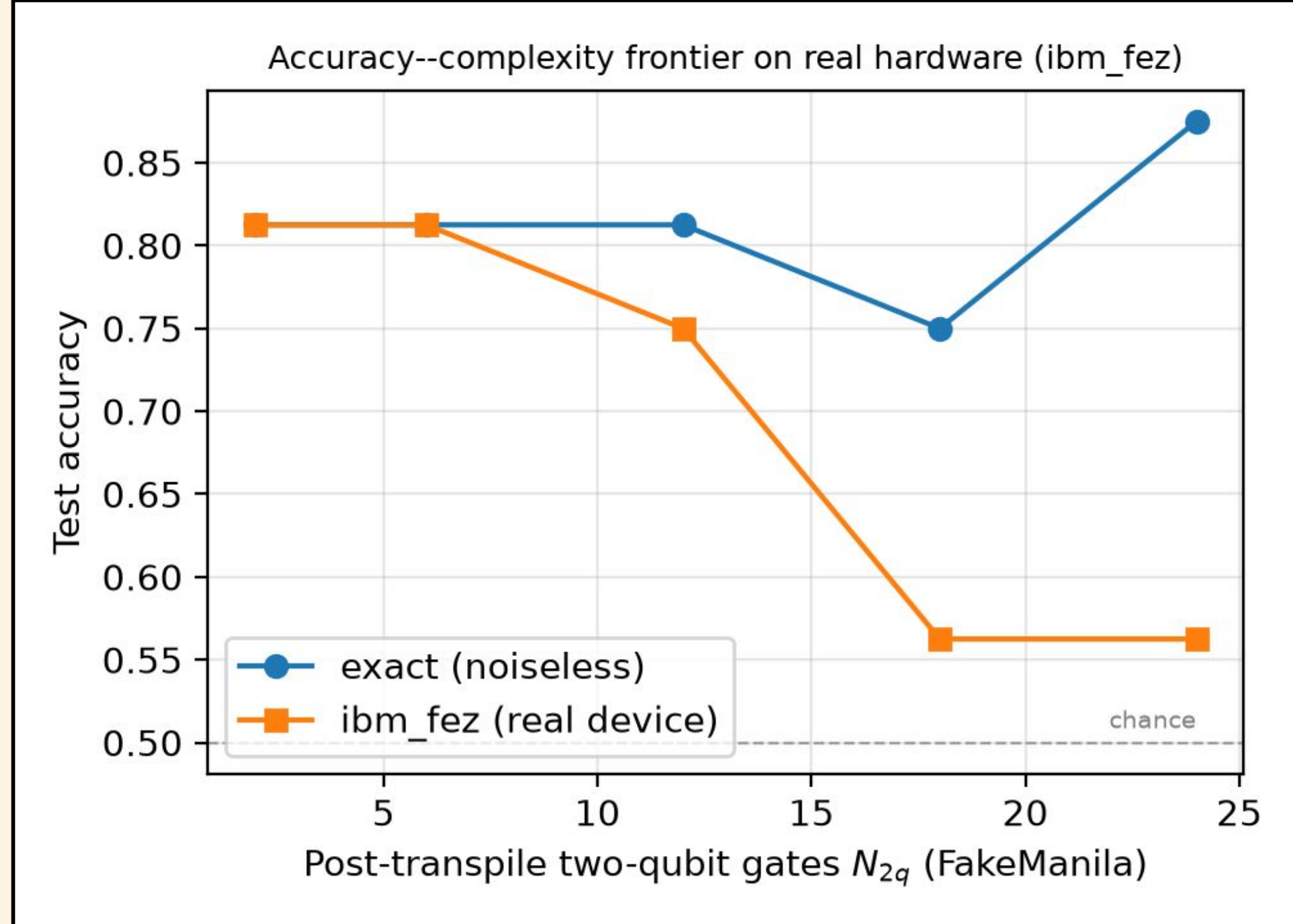
N_{2q}	Learned	Random	Δ
2	0.847	0.742	+0.104
6	0.927	0.814	+0.112
10	0.913	0.794	+0.119
14	0.917	0.826	+0.091
18	0.900	0.809	+0.091
22	0.897	0.836	+0.061

Structure, visualized — learned vs. random



- At the **same N_{2q}** , greedy pruning keeps task-relevant interactions while random masks discard them — the **pattern** of retained entanglers, not the count, drives accuracy.
- Matched **N_{2q}** does not mean matched capacity: two masks with the same gate count can preserve very different interaction patterns.
- Greedy pruning keeps the (XX/YY/ZZ) terms that support the checkerboard rule, while random masks often remove useful feature-coupling structure.

6 Result 3 — Hardware flips the frontier



- Trained circuits run on physical IBM **ibm_fez** (156-qubit Heron), hard task, 5 device budgets, **16 test stimuli @ 2048 shots** (endpoints: 40 stimuli @ 4096 shots).
- The 5-seed exact frontier is flat; **on this device accuracy falls** with N_{2q} : 0.81 @ $N_{2q}=2,6$ → 0.75 @ 12 → **collapses to 0.56** (near chance) @ $N_{2q}=18,24$.
- Endpoint confirmation: **full circuit 0.60** (sim 0.95) vs **compressed 0.83** (sim 0.93).

Single-seed demonstration ($n = 16 / 40$, defaults-only mitigation); error bars are wide, but the direction is consistent across budgets.

TAKEAWAY

On this NISQ device, hardware-aware compression is not merely economical but **necessary** — surplus entangling capacity is actively harmful, and task-aware pruning is what makes the learner usable at all.

7 Robustness — not an artifact

- Device noise model** (FakeManila, 4096 shots, 5 seeds): noisy accuracy tracks the noiseless frontier within a few % at every budget → savings genuine net of noise.
- Density-matrix model** (genuine mixed state, non-unital channel): reproduces every finding — flat frontier, collapse at $N_{2q}=0$, hard peak **0.88 @ 6 vs 0.87 @ 24**, learned > random (mean **+0.043**).
- Classical baselines** (identical splits, 5 seeds): logistic regression at chance on every level; RBF-SVM & MLP fall monotonically easy→hard. Best compressed QBL: **easy 0.970, medium 0.973, hard 0.927**.

8 Key takeaways

- Most entanglement is **redundant** — sheds **~2/3** of two-qubit gates for **free**, **>90%** at modest cost.
- Entanglement is **necessary** — removing all of it collapses the learner to chance: a concrete **capacity boundary**.
- Structure beats count** — learned masks beat random at every matched budget.
- On real hardware, compression is what makes the learner work at all.**

References · Acknowledgments · Links

Future work: Multi-qubit belief registers & higher-dim stimuli; richer non-unital evidence channels; hardware-aware objectives weighting depth, coherence & native fidelity; measured device error rates fed back into the compression objective.

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Acknowledgments: Device runs used IBM Quantum services. The author thanks Duke University Pratt School of Engineering, Dr. Shahid Parvez & Dr. Bushra Anjum for feedback & support.

AI Use Disclosure: The author used Claude (Anthropic) for proofreading and language editing, and for assistance generating the figures. All content was reviewed and edited by the author, who takes full responsibility for the final work.



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zoraizmohammad/
qb-learner-compression
Mohammad Zoraiz
mz248@duke.edu